

XRPL Validator — Daily Engineering Report • 2026-03-29

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 5 commits (xrpl-validator 5)

Engineering changes

- **feat: 47 consecutive state hash matches — validator independently verifies XRPL** (b7d73cf , xrpl-validator)
Major fixes that achieved consistent state hash matching against the network: - Sort transactions by TransactionIndex (execution order != array order) Root cause: OfferCancel at exec#19 appeared at array#46, OfferCreate at exec#21 at array#15 — processing in wrong order caused silent state drift - Add FeeSettings singleton to force-fetch list (4 protocol singletons total) - Handle entryNotFound for modified objects ...
- **fix: LedgerHashes sub-page key computed dynamically (seq/65536)** (cd280e4 , xrpl-validator) The XRPL state tree has a LedgerHashes sub-page that changes every ledger (new hash appended). Key = SHA512Half(0x0073 || uint32_be(seq/65536)). This object was never in AffectedNodes and wasn't being fetched, causing a persistent 32-byte drift after ~30-200 ledgers.
- **perf: branch-cached hash computation + skip second download** (71c6f35 , xrpl-validator)
- Cache 16 root-level branch hashes, only recompute dirty branches - Fast path now marks dirty branches (was the cause of optimized-mode drift) - Skip second download entirely — ws_sync starts from pinned ledger - Skip-ahead when >50 ledgers behind (prevents rippled pruning stalls) - Dynamic LedgerHashes sub-page key: SHA512Half(0x0073 || seq/65536)
- **perf: single-pass merge + spawn_blocking — every ledger verified, feed stays smooth** (5cf2901 , xrpl-validator)
- Replace O(k*n) Vec inserts with O(n) single-pass merge: update drops from 6-9s to 0.4s - Total hash time 2.8s — under the 3.5s ledger interval, now verifying every ledger - Move update_and_hash to spawn_blocking: SSE feed, peers, and web UI no longer freeze - Faster metadata retry bail-out: 3 attempts then break batch (prevents stale-ledger stall) - Partial failure tolerance: apply successful fetches even if some ...

Docs and operations

- **license: replace MIT with proprietary — permission required for use** (5ec2dfe , xrpl-validator)

Appendix — commits

Time	Repo	Commit	Subject
01:32	xrpl-validator	b7d73cf	feat: 47 consecutive state hash matches — validator independently verifies XRPL
10:49	xrpl-validator	cd280e4	fix: LedgerHashes sub-page key computed dynamically (seq/65536)
10:59	xrpl-validator	5ec2dfe	license: replace MIT with proprietary — permission required for use
15:34	xrpl-validator	71c6f35	perf: branch-cached hash computation + skip second download
23:19	xrpl-validator	5cf2901	perf: single-pass merge + spawn_blocking — every ledger verified, feed stays smooth